# Singletons, binary unions and ordered pairs exist in Z

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*Editorial scope.* Statement and complete author proof are included verbatim as the span lines around ollabel prop:pairsconsequences of the pinned original source; no text is rewritten or paraphrased.

*Allowed standard prerequisites (editorial).*

- the Pairing axiom of Z
- the Union axiom of Z
- the Extensionality axiom for set equality

**Proposition 0.1.** *For any sets $a$ and $b$, the following sets exist:*

1. $\{a\}$
2. $a \cup b$
3. $\langle a, b\rangle$

*Proof.* 1. By Pairs, $\{a, a\}$ exists, which is $\{a\}$ by Extensionality.
2. By Pairs, $\{a, b\}$ exists. Now $a \cup b = \bigcup\{a, b\}$ exists by Union.
3. By 1, $\{a\}$ exists. By Pairs, $\{a, b\}$ exists. Now $\{\{a\}, \{a, b\}\} = \langle a, b\rangle$ exists, by Pairs again. □

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